

Spatial Statistics in Epidemiology and Public Health

Lecture 7: Point Processes

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Preliminaries

Random patterns

Heterogeneous Poisson process

Estimating intensities: Where events cluster

Spatial relative risk for the gravesite data

Case study: Sea turtle nesting

Juno Beach, Florida: What is our question?

Data: What did we see?

Comparing Nesting patterns: What did we do?

Second order properties: How events cluster

K functions

Monte Carlo envelopes

References

- ▶ Baddeley, A., Rubak, E., and Turner, R. (2015) *Spatial Point Patterns: Methodology and Applications in R*. Boca Raton, FL: CRC/Chapman & Hall.
- ▶ Diggle, P.J. (1983) *Statistical Analysis of Spatial Point Patterns*. London: Academic Press.
- ▶ Diggle, P.J. (2013) *Statistical Analysis of Spatial and Spatio-Temporal Point Patterns, Third Edition* CRC/Chapman & Hall.
- ▶ Waller and Gotway (2004, Chapter 5) *Applied Spatial Statistics for Public Health Data*. New York: Wiley.
- ▶ Møller, J. and Waagepetersen (2004) *Statistical Inference and Simulation for Spatial Point Processes*. Boca Raton, FL: CRC/Chapman & Hall.

Goals

- ▶ Describe basic types of spatial point patterns.
- ▶ Introduce mathematical models for random patterns (stochastic processes) of point-location events.
- ▶ Introduce analytic methods for describing *where* and *how* patterns occur in observed collections of events.
- ▶ We model the *location* of each event as a random variable in space.
- ▶ *NOTE:* These probability models often motivate the model structures we use for disease mapping, spatial (count) regression, etc.

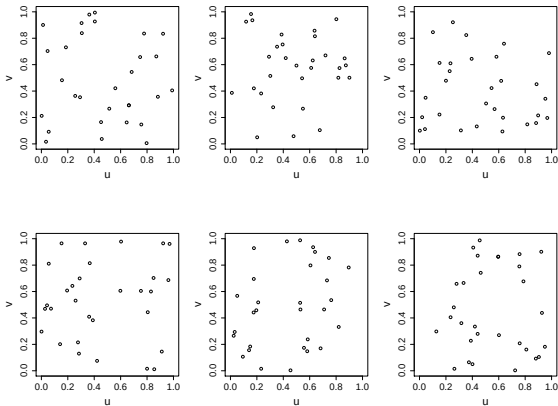
Terminology

- ▶ *Realization*: An observed set of event locations (a data set).
- ▶ *Point*: Where an event *could* occur.
- ▶ *Event*: Where an event *did* occur.

Complete Spatial Randomness (CSR)

- ▶ Start with a model of “lack of pattern”.
- ▶ Events equally likely to occur anywhere in the study area (uniform distribution).
- ▶ Event locations independent of each other.

Six realizations of CSR



CSR as a boundary condition

CSR serves as a boundary between:

- ▶ Patterns that are more “clustered” than CSR.
- ▶ Patterns that are more “regular” than CSR.

Estimating intensities: Where events cluster

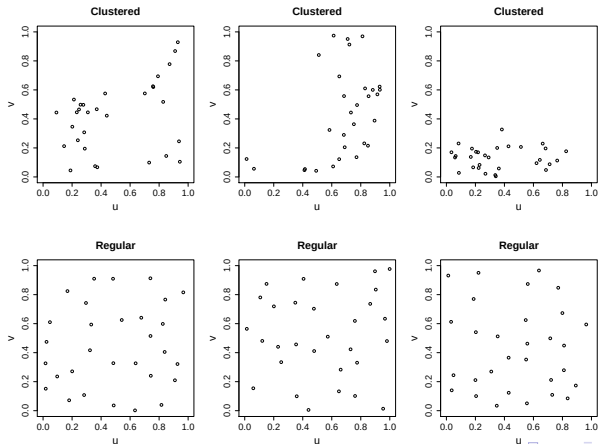
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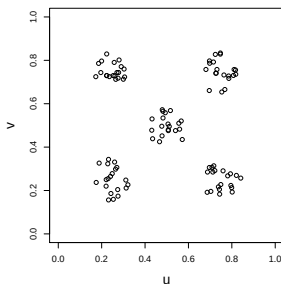
Heterogeneous Poisson process

Too Clustered (top), Too Regular (bottom)

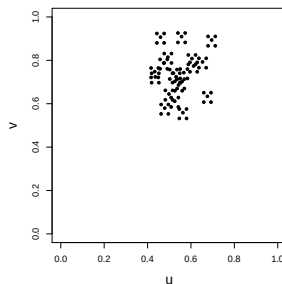


Clusters of regular patterns/Regular clusters

Regular pattern of clusters



Cluster of regular patterns



Spatial Point Processes

- ▶ Mathematically, we treat our point patterns as realizations of a *spatial stochastic process*.
- ▶ A stochastic process is a collection of random variables $X_1, X_2, \dots, X_N$.
- ▶ Examples: Number of people in line at grocery store.
- ▶ For us, each random variable represents an event location.

CSR as a Stochastic Process

Let $N(A)$ = number of events observed in region A , and λ = a positive constant.

A *homogenous spatial Poisson point process* is defined by:

- (a) $N(A) \sim \text{Pois}(\lambda|A|)$
- (b) given $N(A) = n$, the locations of the events are uniformly distributed over A .

λ is the *intensity* of the process (mean number of events expected per unit area).

Is this CSR?

- ▶ Criteria (a) and (b) give a “recipe” for simulating realizations of this process:
 - * Generate a Poisson random number of events.
 - * Distribute that many events uniformly across the study area.
`runif(n,min(x),max(x))`
`runif(n,min(y),max(y))`
- ▶ If we can simulate data sets from CSR, maybe we can see whether our data are consistent or inconsistent with CSR...

Monte Carlo testing

Let T = a random variable representing a test statistic (some numerical summary of the observed data).

What is the distribution of T under H_0 ?

1. Calculate t_1 for your data (the *observed value* of the statistic).
2. simulate $t_2, \dots, t_m$ under H_0 , i.e., what the test statistic would look like under H_0 .
3. $\text{p.value} = \frac{\text{rank of } t_1}{m}$ (where does t_1 fall in the histogram under H_0 ?).

M.C. tests useful in spatial statistics since we can simulate spatial patterns and calculate the statistics.

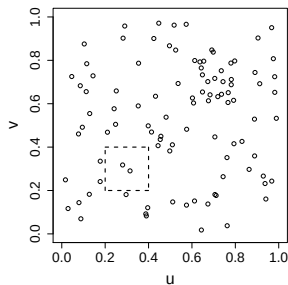
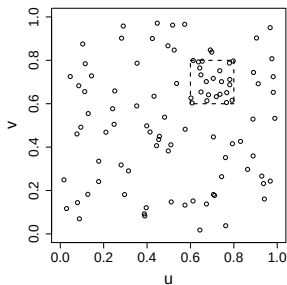
Example: e.g. 592 leukemia cases in ~ 790 regions

Moving beyond CSR

CSR:

1. is the “white noise” of spatial point processes.
2. characterizes the absence of structure (signal) in data.
3. often the null hypothesis in statistical tests to determine if there is clustering in an observed point pattern.
4. not as useful in public health? Why not?

Heterogeneous population density



Heterogeneous Poisson Process

What if λ , the *intensity* of the process (mean number of events expected per unit area), varies by location?

1. $N(A) = \text{Pois} \left(\int_{(\mathbf{s}) \in A} \lambda(\mathbf{s}) d\mathbf{s} \right)$
 $(|A| = \int_{(\mathbf{s}) \in A} d\mathbf{s})$
2. Given $N(A) = n$, events distributed in A as an independent sample from a distribution on A with p.d.f. proportional to $\lambda(\mathbf{s})$.

We still have *counts from areas* $\sim \text{Poisson}$ and *events are distributed proportional to the intensity*.

Estimating intensities: Where events cluster

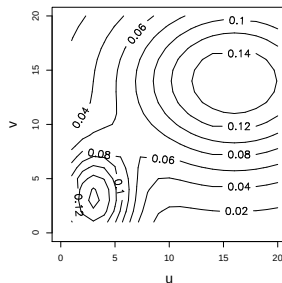
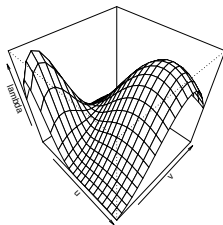
Spatial relative risk for the gravesite data

Case study: Sea turtle nesting

Second order properties: How events cluster

Heterogeneous Poisson process

Example intensity function



Estimating intensities: Where events cluster

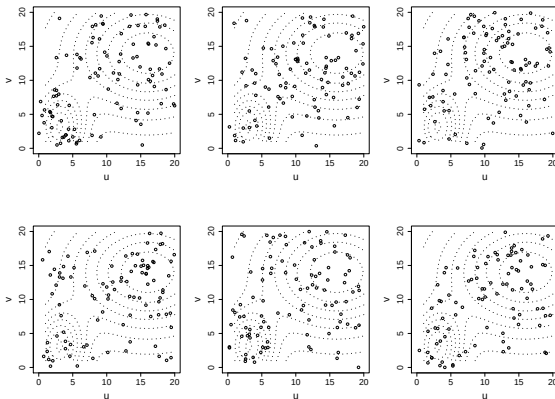
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Six realizations



IMPORTANT FACT!

Without additional information, no analysis can differentiate between:

1. *independent* events in a *heterogeneous (non-stationary)* environment
2. *dependent* events in a *homogeneous (stationary)* environment

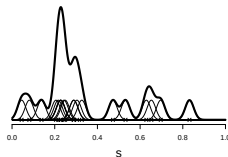
How do we estimate intensities?

Kernel estimators provide a natural approach (Silverman (1986), Wand and Jones (1995, KernSmooth), Baddeley, Rubak, and Turner, (2015,spatstat)).

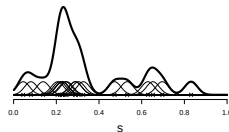
Main idea: Put a little “kernel” of density at each data point, then sum to give the estimate of the overall density function.

Kernels and bandwidths

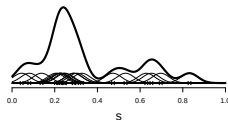
Kernel variance = 0.02



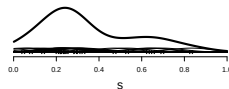
Kernel variance = 0.03



Kernel variance = 0.04



Kernel variance = 0.1



Kernel estimation in R

base

- ▶ `density()` one-dimensional kernel

`library(KernSmooth)`

- ▶ `bkde2D(x, bandwidth, gridsize=c(51, 51),
range.x=<<see below>>, truncate=TRUE)` block kernel
density estimation

`library(splancs)`

- ▶ `kernel2d(pts,poly,h0,nx=20,
ny=20,kernel='quartic')`

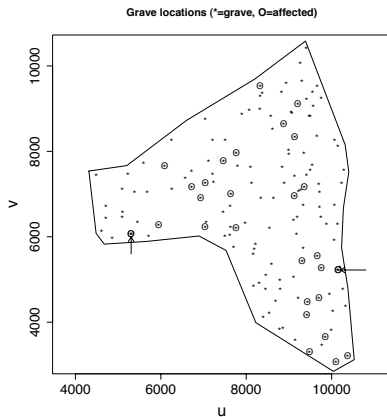
`library(spatstat)`

- ▶ `ksmooth.ppp(x, sigma, weights, edge=TRUE)`

Data Break: Early Medieval Grave Sites

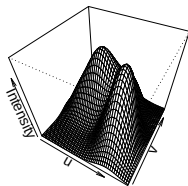
- ▶ Alt and Vach (1991). (Data from Richard Wright Emeritus Professor, School of Archaeology, University of Sydney.)
- ▶ Archeological dig in Neresheim, Baden-Württemberg, Germany.
- ▶ Question: are graves placed according to family units?
- ▶ 143 grave sites, 30 with missing or reduced wisdom teeth.
- ▶ Could intensity estimates for grave sites with and without wisdom teeth help answer this question?

Plot of the data

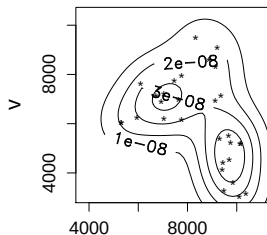


Case intensity

Estimated intensity function

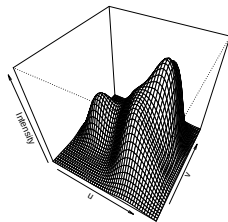


Affected grave locations

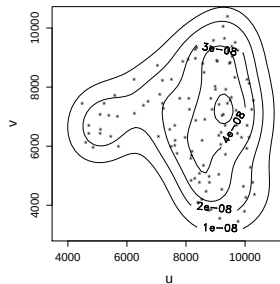


Control intensity

Estimated intensity function



Non-affected grave locations



What we have/don't have

- ▶ Kernel estimates suggest *where* there might be differences.
- ▶ No significance testing (yet!)

Can we compare patterns spatially?

- ▶ Of course!
- ▶ How can we compare first order intensities?
- ▶ What questions would that answer?
- ▶ Do intensities differ between two observed patterns?
- ▶ (Spatial version) *Where* do intensities differ between two observed patterns?

Spatial relative risk

- ▶ If $\lambda_{case}(s)$ is the intensity of cases, and $\lambda_{control}(s)$ is the intensity of controls over the same area.
- ▶ Intensities are the expected number of events at location s .
- ▶ Intensities are proportional to the probability of an event at location s .
- ▶ What if we consider $\frac{\lambda_{case}(s) / \int \lambda_{case}(s) ds}{\lambda_{control}(s) / \int \lambda_{control}(s) ds}$ over the study area?
- ▶ This ratio is the *relative risk* of a case at location s .

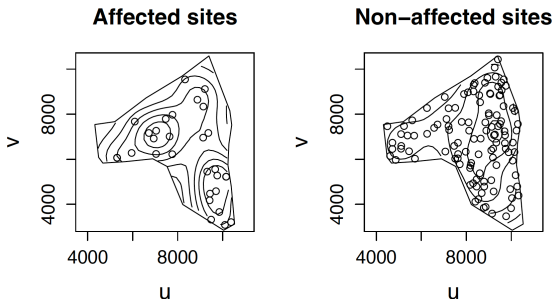
Comparing intensities

- ▶ We want to compare the spatial pattern of cases to the spatial pattern of controls.
- ▶ So far we have the intensity for cases (affected sites) and for controls (non-affected sites).
- ▶ Estimate these using *same grid of locations* and *same bandwidth* (why?).
- ▶ At each grid point, take the ratio of the intensity estimates (at that location).
- ▶ *Random labelling* null hypothesis.

Random labelling

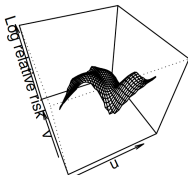
- ▶ CSR was our null hypothesis for one observed pattern.
- ▶ But our question here is more specific. Are the case-control intensities the same, and if they differ *where* do they differ?
- ▶ How about: H_0 : The probability that any gravesite is affected is the same.
- ▶ Select 30 locations to be cases, the rest as controls, calculate the two intensities (same grid, same bandwidth), calculate ratio at each grid point.
- ▶ Repeat this many times and make a histogram of the ratio at each gridpoint.
- ▶ See where observed ratio falls in histogram at each gridpoint.
- ▶ Indicate grid points where observed ratio is either over 97.5th percentile or under 2.5th percentile.

Dental intensities (code in lab for this afternoon)

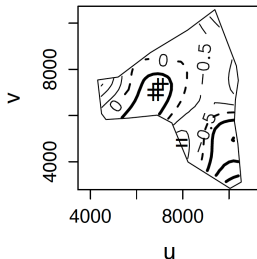


Dental Relative Risk Surface

Log relative risk surface

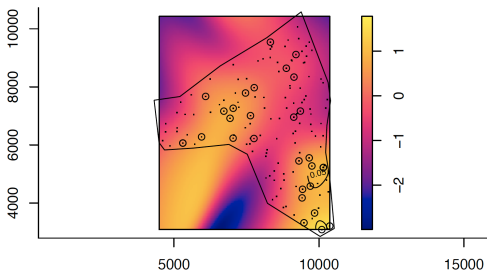


Gaussian kernel, Bandwidth = 700



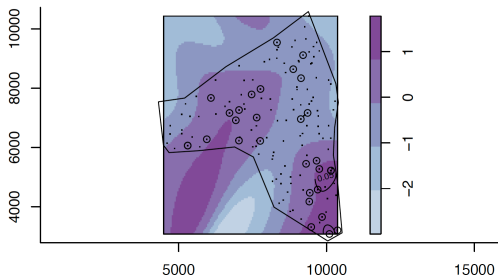
Dental Relative Risk Surface (sparr, default color)

Sparr relative risk surface, bandwidth = 700



Dental Relative Risk Surface (sparr, blue to purple)

RR surface, bw = 700 Blue to Purple



Summary

- ▶ Thoughts? Questions?
- ▶ Color?
- ▶ What happens outside of the polygon?
- ▶ Where are there differences?
- ▶ Are they statistically important?

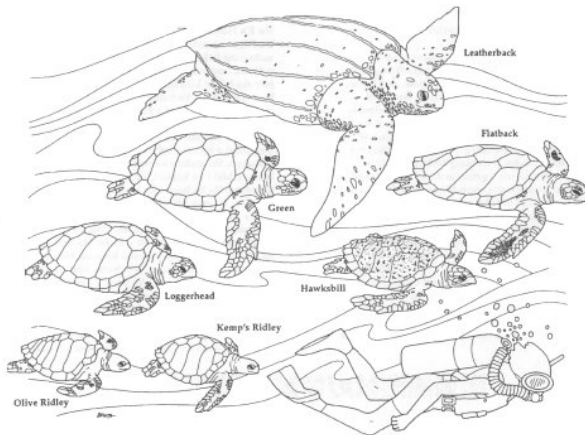
Sea turtle biology

- ▶ Existed (at least) since Jurassic period (~ 200 -120 mya).
- ▶ Air breathing reptile.
- ▶ Spends life in ocean, returns to natal beach to lay eggs.
- ▶ Lays eggs on land (~ 100 eggs per clutch, buried in sand).
- ▶ Hatchlings appear *en masse*, head to the ocean.

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Sea Turtle Species



Consider the hatchling

- ▶ Life fraught with peril.
- ▶ On menu for crabs, gulls, etc.
- ▶ Goal 1: Get to the ocean. Follow the light!

Sea Turtles and Humans: Regulatory implications

- ▶ All species “threatened” or “endangered” .
- ▶ Regulations in place for fisheries (TEDs), beach-front lighting, beach-front development.
- ▶ East coast of Florida (USA), prime nesting site for loggerhead turtles.
- ▶ Also popular among green turtles (biannual cycle).
- ▶ We consider impact of two different construction projects on Juno Beach, Florida.

Two construction projects

- ▶ Construction of 990-foot fishing pier in 1998.
- ▶ Environmental impact statement approved if collect data to determine impact on sea turtle nesting.
- ▶ Beach nourishment project 2000.
- ▶ Dredge up sand offshore and rebuild beach.
- ▶ Past studies on impact of beach nourishment on turtle nesting (reduced for 2 nesting seasons, off-shore slope impact?)

“Standard” approach

- ▶ Patrol beach each morning during nesting season.
- ▶ Count trackways in each Index Nesting Beach Zone.
- ▶ Compare distribution of counts between years.

Juno Beach data

- ▶ Marinelife Center of Juno Beach volunteers patrol beach each morning.
- ▶ Differential GPS coordinates for apex of each “crawl”.
- ▶ Store: Date. Nest? Species. Predation?
- ▶ Approximately 10,000 points per year.

Juno Beach, Florida: What is our question?

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Juno Beach Data

Juno Beach Turtle Nesting Data 1998-2000

Layers:

Photographs

GPS Emergences

Index Nesting Beach

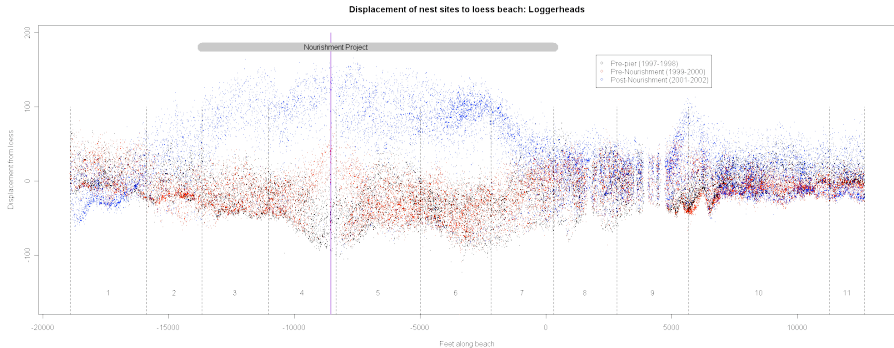
□ (INB) Zones



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Loggerhead Displacement from Beach



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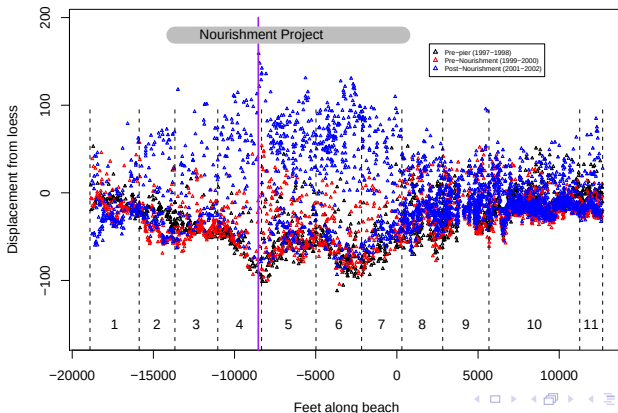
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Displacement of Green Emergences

Displacement of nest sites to loess beach: Greens



“Linearizing” the beach

- ▶ While we have sub-meter accurate GPS locations, depth on beach of secondary interest.
- ▶ We want to know *where* differences in nesting pattern occur and whether these observed differences are significant.
- ▶ Treat emergence locations as realization of a spatial point process in one dimension (events on a line).
- ▶ Ignore depth on beach by projecting all points to loess curve representing beach.
- ▶ Three time periods of interest: Pre-pier (1997-1998), pre-nourishment (post-pier) (1999-2000), post-nourishment (2001-2002).

Kernel estimates for each time period

- ▶ Let $\lambda(\mathbf{s})$ denote *intensity function* of point process (mean number of events per unit distance).
- ▶ Intensity proportional to pdf.
- ▶ Data: locations $\mathbf{s}_1, \dots, \mathbf{s}_n$.
- ▶ Estimate via kernel estimation:
$$\hat{\lambda}(\mathbf{s}) = \frac{1}{nb} \sum_{i=1}^n \text{Kern} \left[\frac{(\mathbf{s} - \mathbf{s}_i)}{b} \right]$$
- ▶ Kern($\cdot$) is kernel function, b bandwidth (governing smoothness).

Ratio of kernel estimates

- ▶ Kelsall and Diggle (1995, *Stat in Med*) consider ratio of two intensity estimates as spatial measure of relative risk.
- ▶ Suppose we have n_1 “before” events, n_2 “after” events.
- ▶ $\lambda_1(\mathbf{s})$, $\lambda_2(\mathbf{s})$ associated before/after intensity functions.
- ▶ Conditional on n_1 and n_2 , data equivalent to two random samples from pdfs:
$$f_1(\mathbf{s}) = \lambda_1(\mathbf{s}) / \int \lambda_1(\mathbf{u}) d\mathbf{u} \text{ and}$$
$$f_2(\mathbf{s}) = \lambda_2(\mathbf{s}) / \int \lambda_2(\mathbf{u}) d\mathbf{u}.$$

Ratio of kernel estimates

- ▶ Consider *log relative risk*

$$r(\mathbf{s}) = \log \frac{f_1(\mathbf{s})}{f_2(\mathbf{s})}.$$

- ▶ Some algebra yields:

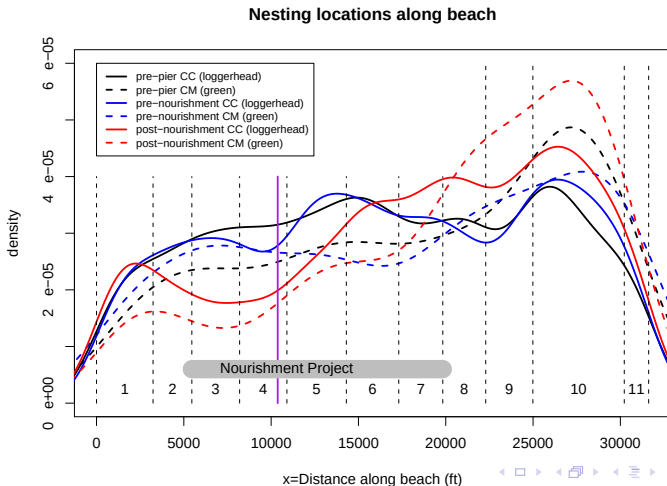
$$r(\mathbf{s}) = \log \left[\frac{\lambda_1(\mathbf{s})}{\lambda_2(\mathbf{s})} \right] - \log \left[\frac{\int \lambda_1(\mathbf{u}) d\mathbf{u}}{\int \lambda_2(\mathbf{u}) d\mathbf{u}} \right].$$

- ▶ $H_0 : r(\mathbf{s}) = 0$ for all $\mathbf{s}$.
- ▶ How do we test this?

Monte Carlo inference

- ▶ Conditional on all $n_1 + n_2$ locations, randomize n_1 to “before”, n_2 to “after”.
- ▶ Calculate $\hat{f}^*(\mathbf{s})$ and $\hat{g}^*(\mathbf{s})$, get ratio $r^*(\mathbf{s})$.
- ▶ Repeat many (999) times.
- ▶ Calculate pointwise 95% tolerance region.

Densities

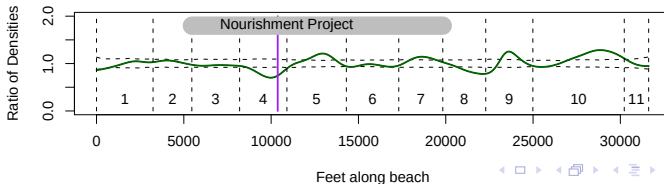
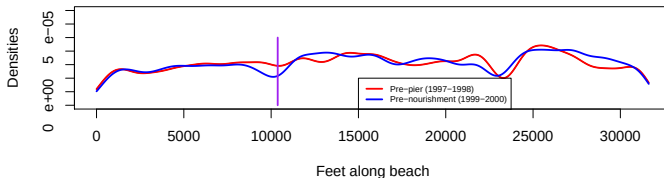


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Pre- vs. post-pier, Loggerheads

Pre-pier vs. pre-nourishment, Loggerheads



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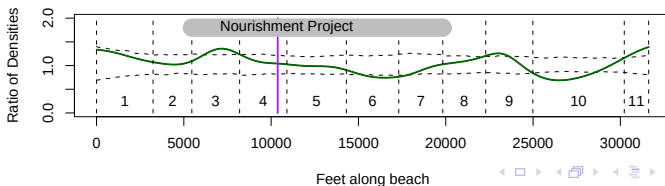
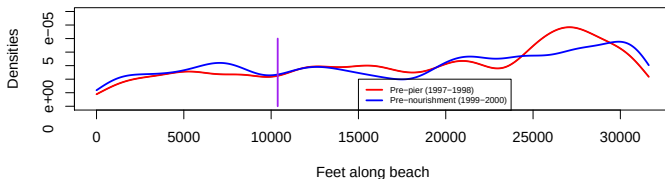
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Pre- vs. post-pier, Greens

Pre-pier vs. pre-nourishment, Greens



Differences

- ▶ Green densities smoother (wider bandwidth due to smaller sample size).
- ▶ Clear impact of pier on loggerheads (but very local).
- ▶ Conservation strategy? Maintain protected areas nearby.
- ▶ Note: impact on *all* crawls (nesting and non-nesting).

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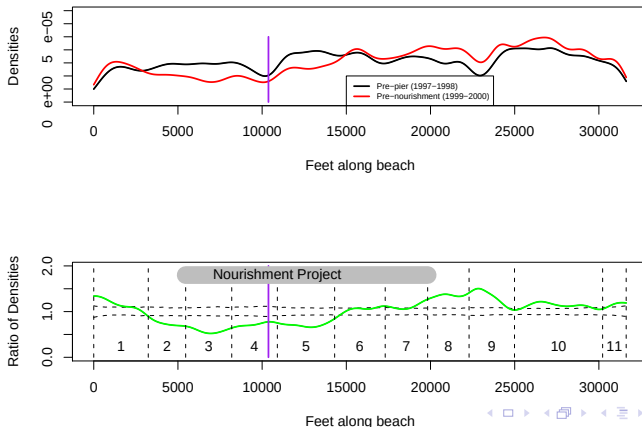
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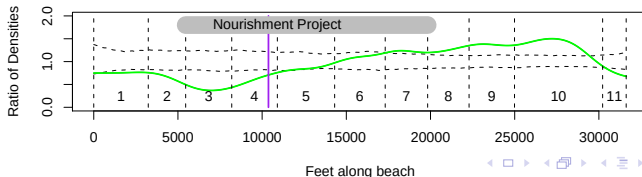
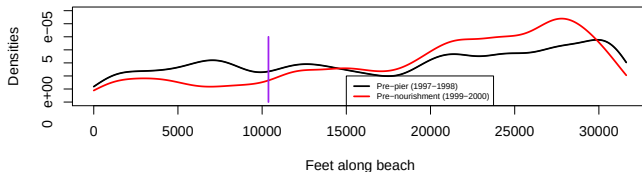
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Pre- vs. post-nourishment, Greens

Pre-nourishment vs. post-nourishment, Greens



Differences

- ▶ Reduced nesting in nourishment zone, increases just to south.
- ▶ Offshore current suggests approach from north (left on plot).
- ▶ Impact on *all* crawls (nesting and non-nesting).
- ▶ Suggests impact of nourishment *before* turtle emerges. Slope?

Conclusions

Pairwise comparisons suggest:

- ▶ Local impact of pier for loggerheads, but not for greens.
- ▶ Shift to south in emergences (nesting and non-nesting) due to nourishment for both loggerheads and greens.

First and Second Order Properties

- ▶ What else can we explore in observed point patterns?
- ▶ The intensity function describes *where* events cluster.
- ▶ What about *second order* properties relating to covariance/correlation between event locations (if events non independent...)?
- ▶ This can tell us *how* events cluster.

Ripley's K function

Ripley (1976, 1977 introduced) the *reduced second moment measure* or *K function*

$$K(h) = \frac{E[\# \text{ events within } h \text{ of a randomly chosen event}]}{\lambda},$$

for any positive *spatial lag* h .

- ▶ Under CSR, $K(h) = \pi h^2$ (area of circle of with radius h).
- ▶ Clustered? $K(h) > \pi h^2$.
- ▶ Regular? $K(h) < \pi h^2$.

Calculating $K(h)$ in R

```
library(splancs)
```

- ▶ `khat(pts,poly,s,newstyle=FALSE)`
- ▶ `poly` defines polygon boundary (important!!!).

```
library(spatstat)
```

- ▶ `Kest(X, r, correction=c("border", "isotropic", "Ripley", "translate"))`
- ▶ Boundary part of `X` (point process “object”).

Plots with $K(h)$

- ▶ Plotting $(h, K(h))$ for CSR is a parabola.
- ▶ $K(h) = \pi h^2$ implies

$$\left(\frac{K(h)}{\pi} \right)^{1/2} = h.$$

- ▶ Besag (1977) suggests plotting

$$h \text{ versus } \hat{L}(h)$$

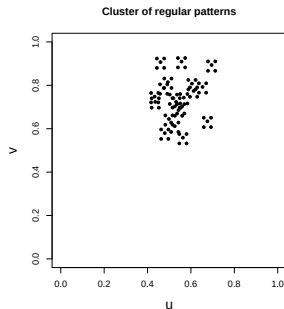
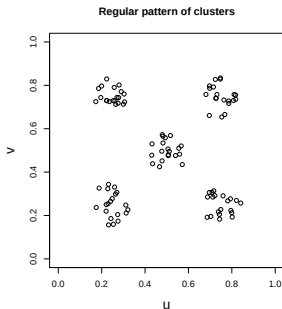
where

$$\hat{L}(h) = \left(\frac{\hat{K}_{ec}(h)}{\pi} \right)^{1/2} - h$$

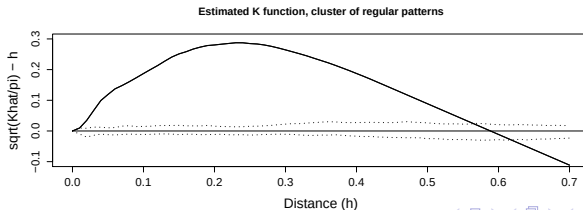
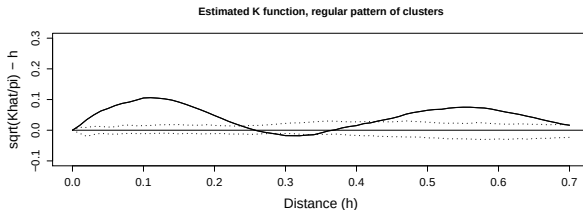
Monte Carlo Variability and Envelopes

- ▶ Observe $\hat{K}(h)$ from data.
- ▶ Simulate a realization of events from CSR.
- ▶ Find $\hat{K}(h)$ for the simulated data.
- ▶ Repeat simulations many times.
- ▶ Create simulation “envelopes” from simulation-based $\hat{K}(h)$ ’s.

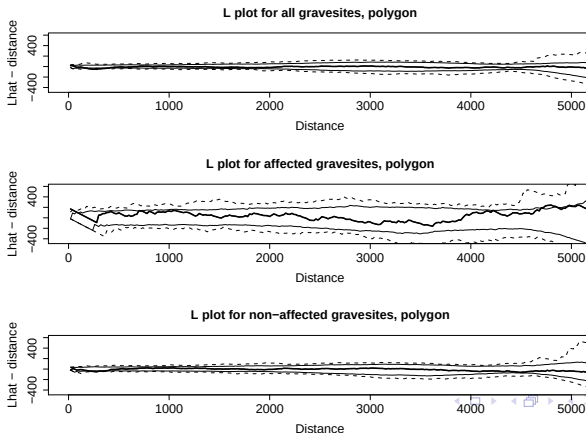
Recall: Clusters of regular patterns/Regular clusters



K functions for regular clusters and clusters of regularity



Data break: Medieval graves: K functions with polygon adjustment



Clustering?

- ▶ Clustering of cases at very shortest distances.
- ▶ Likely due to two coincident-pair sites (both cases in both pairs).
- ▶ Envelopes based on random samples of 30 “cases” from set of 143 locations (random labelling null hypothesis).

Notes

- ▶ First and second moments do not uniquely define a distribution, and $\lambda(\mathbf{s})$ and $K(h)$ do not *uniquely* define a spatial point pattern (Baddeley and Silverman 1984, and in Section 5.3.4).
- ▶ Analyses based on $\lambda(\mathbf{s})$ typically assume independent events.
- ▶ Analyses based on $K(h)$ typically assume a stationary process (with constant λ).
- ▶ Remember IMPORTANT FACT! above.

What questions can we answer?

- ▶ Are events uniformly distributed in space?
 - ▶ Test CSR.
- ▶ If not, where are events more or less likely?
 - ▶ Intensity estimation.
- ▶ *Where* do two spatial patterns differ?
 - ▶ Ratio of intensity functions.
- ▶ Do events tend to occur near other events, and, if so, at what scale?
 - ▶ K functions with Monte Carlo envelopes.
- ▶ *How* do two spatial patterns cluster differently at different scales?
 - ▶ Difference of K functions